Mark Scheme (Results)

January 2023

Pearson BTEC Nationals
In Applied Science (31617H/1B)
Unit 1: Principles and Applications of Science I
Biology

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Unit 1: Principles and Applications of Science I Biology

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Mark grids should be applied positively. Learners must be rewarded for what they have shown they can do rather than be penalised for omissions.
- Examiners should mark according to the mark grid, not according to their perception of where the grade boundaries may lie.
- All marks on the mark grid should be used appropriately.
- All the marks on the mark grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the mark grid.
- Where judgement is required, a mark grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the mark grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The mark grids have been designed to assess learners' work holistically.

Rows in the grids identify the assessment focus/outcome being targeted. When using a mark grid, the 'best fit' approach should be used.

- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band depending on how they have evidenced each of the descriptor bullet points.

Question Number	Answer	Additional Guidance	Mark
1 (a)	B – chloroplast		1
1 (b)	80(S)		1
1 (c)	Any two from: Absorb water (1) Absorb minerals (1) Increase surface area (1) Accept any other valid response.	Accept nutrients, ions Accept named examples	2
1 (d)	Conversion (1) 13 200(μm) Substitution (1) $\frac{13\,200}{66}$ Evaluation (1) 200(x) OR Conversion (1) 0.066(mm) Substitution (1) $\frac{13.2}{0.066}$ Evaluation (1) 200(x)	200(x) alone gains all 3 marks Allow ECF For a power of ten error award 2 marks	3

Question Number	Answer	Additional Guidance	Mark
2 (a)	Vesicle (Containing neurotransmitter/dopamine)		1
2 (b)	R – neurotransmitter / (brain) chemical / molecule S – diffusion T – receptors / (sodium/ ion) channel(s)	Accept phonetic spelling	3
2 (c)(i)	Any one from: Parkinson's disease Depression Accept any other valid response.		1
2 (c)(ii)	B – L-Dopa		1

Question Number	Answer	Additional Guidance	Mark
3(a)	Any two from: Epithelial (cells) (1) Squamous/ flat cells (1) Single layer of cells/ one cell thick / thin wall (1) Tightly packed (cells) (1) Contains elastin (1)	 Accept pavement Accept short diffusion pathway Allow flexible/ elastic	2

3 (b)	Award **one** mark for an identification point and **one** mark for an expansion point. Any of the following can be identification or expansion, depending on how the learner shapes their response. Goblet cells secrete mucus (1) (Mucus) traps pathogens/ unwanted particles (1) Cilia waft / beat / wave (1) (Cilia) removes unwanted substances (out of the lungs) (1)	Ignore lubrication Accept produces Accept named examples, e.g. mucus/ pathogens/ particles/toxins/ debris/virus	4

Question Number	Answer	Additional Guidance	Mark
4 (a) (i)	D – ★		1
4 (a) (ii)	Any two from: Phagosome formed/ enclosed in a vesicle (1) Lysosome (fuses with phagosome) (1) (Lysosome contains) enzymes / lysozyme / chemicals (1) (Enzymes) digest / hydrolyse the pathogen (1)	Award 2 marks for phago-lysosome Accept named chemicals, e.g. toxins, H_2O_2 Accept break down	2

4 (a)(iii)	Award **one** mark for an identification point and **one** mark for an expansion point. Any of the following can be identification or expansion, depending on how the learner shapes their response. (B-lymphocytes) recognise/ respond to/bind to (pathogen) antigens (1) Produce/ secrete antibodies/ antitoxins (1) Antibodies are specific to each antigen (1) Some become memory cells (1)	Reject antibiotics	2

Question number	Indicative content
5	Answers will be credited according to the learner's demonstration of knowledge and understanding of the material, using the indicative content and levels descriptors. The indicative content that follows is not prescriptive. Answers may cover some or all the indicative content, but learners should be rewarded for other relevant answers. **Similarities** Both function to control / regulate cellular activity Both have non-coding regions Both contain DNA condensing and supercoiling DNA double stranded in both In both, genes are in the chromosome(s) Both contain RNA **Differences** Nucleus structure In eukaryotes Large organelle Spherical shaped Membrane bound organelle/surround by nuclear envelope/double membrane Contains more than one chromosome/chromosomes are linear has more non-coding regions than nucleoid DNA forms structure with histones / ref to nucleosomes nucleolus and nucleoplasm **nucleoid structure** In prokaryotes Irregularly shaped region Without a (nuclear) membrane Contains a single chromosome / DNA forms one circular chromosome smaller than a nucleus no histone proteins DNA is compact with NAPs (nucleoid associated proteins / chromosomal architectural proteins) Nucleolus and nucleoplasm are absent

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	**Mark**	**Descriptor**
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates adequate knowledge and understanding of scientific facts / concepts to the given context with generalised comments made. • Generic statements may be presented rather than linkages to the context being made so that lines of reasoning are unsupported or partially supported. • The comparison will contain some similarities and differences showing some structure and coherence.
Level 2	3–4	• Demonstrates good knowledge and understanding by selecting and applying some relevant scientific facts / concepts to provide the comparison being presented. • Lines of argument mostly supported through the application of relevant evidence drawn from the context. • Demonstrate an awareness of both similarities and differences leading to a comparison which has a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of scientific facts / concepts to provide the comparison being presented. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence drawn from the context. • The comparison shows a logical chain of reasoning which is supported throughout by sustained application of relevant evidence.

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